

Research Plan: Trend-Following as Crash Insurance

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Research question

Working title: Is Trend-Following Worth Its Cost as Crash Insurance? Evidence from a Century of Industry Returns

Question: Does a moving-average trend-following rule, applied to survivorship-free U.S. industry portfolios from 1926 to 2025, deliver downside protection that is worth its return drag after costs, risk adjustment, and data-snooping correction? And has its performance relative to buy-and-hold changed since Brock, Lakonishok & LeBaron (1992) was published?

This plan is written and locked before any results are computed. Every test below is pre-specified.

Prior exposure (disclosure): Before writing this plan, the author ran the 50/200-day rule against buy-and-hold in an earlier capstone paper on five large-cap U.S. stocks (AAPL, MSFT, JPM, XOM, JNJ), daily data from 2010 to 2023. The strategy earned lower returns than buy-and-hold with smaller drawdowns, and that result motivated the crash-insurance framing of this paper. The capstone did not use the French industry portfolios, the pre-1993 period, or any test in this plan, but its sample period overlaps the post-1992 subsample in time.

Contribution

The paper reframes the MA rule as a tail-risk hedge and prices that hedge. Prior work mostly asks whether the rule beats buy-and-hold on mean return.

Prior work	What it found	What this paper adds
Brock, Lakonishok & LeBaron (1992)	MA rules profitable on the DJIA, 1897-1986	Survivorship-free cross-section of 49 industries instead of one index
Sullivan, Timmermann & White (1999)	Edge does not survive data-snooping correction out of sample (1987-1996)	30 more years out of sample; SPA test; deflated Sharpe
Han, Yang & Zhou (2013)	MA timing profitable on volatility-sorted portfolios	Industry portfolios; crash-period and utility-based evaluation
Hurst, Ooi & Pedersen (2017)	Trend-following pays in crises across asset classes (futures)	Equity-only evidence across a 90-rule catalog, with formal tail tests
McLean & Pontiff (2016)	Anomaly returns fall after publication	Direct pre/post-1992 test for MA rules

Three contributions:

1. A survivorship-free, century-long test of the 50/200-day rule across 49 U.S. industries.
2. A formal valuation of the rule's downside protection: crash-period returns, CVaR, and certainty-equivalent gains for risk-averse investors.

3. A publication-decay test combined with a data-snooping-robust (SPA) evaluation of a pre-specified 90-rule catalog spanning 8 trend and risk-management families, each tested against a constant-exposure benchmark.

Hypotheses

H1 is the primary hypothesis; H2-H6 are secondary. All tests are two-sided at $\alpha = 0.05$ unless stated otherwise. H4 is one-sided because the crash-insurance hypothesis is directional: the rule is claimed to reduce tail loss, and that direction is fixed here before any data are seen. $\alpha = 0.05$ decides significance; the $t > 3.0$ hurdle of Harvey, Liu & Zhu (2016) is reported only as an extra, stricter check.

#	Null hypothesis (H_0)	Alternative (H_1)	Test
H1 (primary)	$\Delta CE = 0$: the strategy's CRRA certainty equivalent ($\gamma = 5$) equals buy-and-hold's	$\Delta CE \neq 0$	Mean test on monthly utility differences d_m ; studentized stationary block bootstrap
H2	Sharpe of strategy = Sharpe of buy-and-hold	Sharpes differ	Ledoit-Wolf (2008) studentized block-bootstrap Sharpe test
H3	Factor alpha of strategy = 0	$\alpha \neq 0$	Newey-West t-test on FF3/FF5 + momentum regressions
H4	Strategy CVaR(5%) = buy-and-hold CVaR(5%)	Strategy's CVaR loss is smaller	Stationary block bootstrap on monthly returns, one-sided; daily CVaR reported alongside. Max drawdown is reported descriptively, not tested
H5	ΔCE is the same before and after 1992	It changed after publication	Interaction regression on d_m (Newey-West) plus bootstrap of ΔCE post minus pre
H6	No rule in the 90-rule catalog has $\Delta CE > 0$	At least one does	Hansen (2005) SPA test on d_m

Headline metric: CRRA certainty equivalent. CE is the guaranteed return a risk-averse investor would accept instead of the strategy. It is computed on monthly net returns, compounded from daily, so large monthly crashes carry their full weight. With $\gamma = 5$ and M months of returns R_m :

$$CE = \left(\frac{1}{M} \sum_{m=1}^M (1 + R_m)^{1-\gamma} \right)^{\frac{1}{1-\gamma}} - 1, \quad \Delta CE = (1 + CE_S)^{12} - (1 + CE_{BH})^{12}$$

ΔCE is annualized, in percent per year. $\Delta CE > 0$ exactly when the strategy's average utility is higher, so H1 is a test on the mean of the monthly utility difference:

$$d_m = \frac{(1 + R_m^S)^{1-\gamma} - (1 + R_m^{BH})^{1-\gamma}}{1 - \gamma}, \quad H_0 : E[d_m] = 0$$

Mean-variance CE = $12 \cdot \text{mean}(R_m) - (\gamma/2) \cdot 12 \cdot \text{var}(R_m)$, computed on monthly net returns in decimals at each γ used for CRRA CE (2, 5, 10). ΔMV -CE is strategy minus buy-and-hold, reported in percent per year.

The Sharpe ratio in H2 uses daily excess returns over the actual daily T-bill rate, not a constant:

$$SR = \frac{\overline{r_t - r_{f,t}}}{\text{sd}(r_t - r_{f,t})} \sqrt{252}, \quad H_0 : SR_S = SR_{BH}$$

The H5 regression, where $D_{post} = 1$ for months from January 1993 onward:

$$d_m = a + b D_{post,m} + \varepsilon_m, \quad H_0 : b = 0$$

Data

The main sample is the Kenneth French 49 Industry Portfolios, daily, July 1926 to December 2025. These are built from CRSP and include delisted firms, so the sample is survivorship-free by construction.

Dataset	Source	Period	Role
49 Industry Portfolios, daily, value-weighted	French Data Library	1926-07 to 2025-12	Main sample
49 Industry Portfolios, daily, equal-weighted	French Data Library	1926-07 to 2025-12	Robustness
Fama-French 3 factors + daily risk-free rate	French Data Library	1926-07 to 2025-12	Cash return; H3 factor regressions (full sample)
Fama-French 5 factors, daily	French Data Library	1963-07 to 2025-12	H3 (from 1963)
Momentum factor (UMD), daily	French Data Library	1926-11 to 2025-12	H3
Cboe S&P 500 5% Put Protection Index (PPUT), daily	Cboe	1986-06-30 to 2025-12 (Cboe); values before the index launch are back-tested by Cboe	Insurance comparison
S&P 500 Total Return Index (^SP500TR), daily	yfinance	1988-01-04 to 2025-12 (Yahoo Finance)	Base index for the PPUT comparison (flagships and buy-and-hold)
International country portfolios, monthly, USD	French Data Library	1975 to 2025-12; MSCI data through 2006, Bloomberg from 2007 (French Data Library); start varies by country	International replication
SPY + 9 SPDR sector ETFs (XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLY)	yfinance, adjusted close	1999-01 to 2025-12	Investable robustness check (flagships 1-7)

Handling rules:

- French’s missing-value code (−99.99) is treated as missing. An industry needs 273 trading days of history (the longest rule’s requirement: family 8’s 12-month lookback plus the skipped month) before it enters either the strategy or the buy-and-hold portfolio.
- Returns are converted to a total-return index. Moving averages are computed on that index, so dividends are included.
- The market-wide series for crash definitions is Mkt-RF + RF from the Fama-French 3-factor file.
- Raw downloaded files are saved with their download date and never edited by hand.

Strategy specification

The primary rule is the 50/200-day SMA crossover with a 10 bp one-way cost. It was fixed before this study, carried over from the capstone paper, so it involves no selection on this data.

Signal and execution:

- Signal at the close of day t : long if $SMA_{50} > SMA_{200}$ on the total-return index, otherwise cash.
- The position earns day $t+1$'s return (one-day lag, no look-ahead). Monthly rules (families 2, 4, and 8, plus family 7's trend signal) signal at the last close of the month and trade on the first trading day of the next month.
- In cash, the strategy earns the daily risk-free rate from the French file.
- Long-only. No shorting, no leverage.
- Evaluation for strategy and buy-and-hold starts on the same day: the first day every rule in the catalog has a valid signal. An industry that enters later joins the strategy and buy-and-hold portfolios on the same day.

Portfolio: Equal weight across all industries active that day, rebalanced daily. Equal weighting is primary so no single industry, such as tech, dominates the result. Value weighting across industries is a robustness check. Monthly rebalancing is a robustness check. Per-industry results are also reported.

Costs: One-way cost c is charged on every change in stock weight, as $c \times |w_t - w_{t-1}|$, so fractional-weight strategies pay in proportion. Base case $c = 10$ bp. Sensitivity grid: 0, 5, 10, 25, 50 bp. Because commissions were fixed and higher before May 1975, a robustness run uses 50 bp before 1975-05 and 10 bp after. The break-even cost (where $\Delta CE = 0$) is reported.

SMA rule grid (family 1 of the catalog, 48 rules):

Parameter	Values
Short window (days)	1, 5, 10, 20, 50 (1 = price vs. long MA)
Long window (days)	50, 100, 150, 200, 250
Constraint	short < long
Band filter	0%, 1%: enter when short > long $\times (1 + \text{band})$; exit when short < long $\times (1 - \text{band})$; otherwise hold

That gives 24 window pairs $\times$ 2 bands = 48 rules. The other 42 rules are in the strategy catalog below.

Strategy catalog

The paper compares 8 strategy families, 90 rules in total, all locked now. Each family has one pre-specified flagship rule for the main comparison; every rule enters the SPA test.

#	Family	Rule	Locked grid	Rules	Flagship
1	SMA crossover	Long if short SMA > long SMA (daily)	See SMA grid above	48	50/200, no band
2	Price vs. monthly MA (Faber 2007)	At month-end, long if price > N-month SMA of month-end prices	N = 6, 8, 10, 12 months	4	10 months
3	EMA crossover	Long if short EMA > long EMA (daily)	Short span 10, 20, 50; long span 100, 150, 200	9	50/200
4	Time-series momentum (Moskowitz, Ooi & Pedersen 2012)	At month-end, long if past L-month return > T-bill return over the same L months	L = 1, 3, 6, 9, 12 months	5	12 months
5	Channel breakout (Brock et al. 1992)	Enter on close > prior N-day high $\times (1 + \text{band})$; exit on close < prior N-day low $\times (1 - \text{band})$; otherwise keep the current position	N = 50, 100, 150, 200, 250; band 0%, 1%	10	N = 200, no band

#	Family	Rule	Locked grid	Rules	Flagship
6	Volatility targeting (adapted from Moreira & Muir 2017)	Weight = $\min(\sigma^* / \hat{\sigma}, 1)$; $\hat{\sigma}$ = annualized realized vol (daily standard deviation $\times \sqrt{252}$) over past V days; weight = 1 if $\hat{\sigma} = 0$; rest in T-bills; rebalanced at month-end or when the target weight moves more than 5 percentage points	V = 21, 63, 126 days; $\sigma^* = 10\%$, 15% annual	6	V = 63, $\sigma^* = 15\%$
7	Trend + vol targeting	Family 4 signal $\times$ family 6 weight, same rebalancing rule as family 6	L = 6, 12 months; V = 63 days; $\sigma^* = 10\%$, 15%	4	L = 12, $\sigma^* = 15\%$
8	Industry momentum (Moskowitz & Grinblatt 1999)	At month-end, hold top K industries by J-month return, skipping the latest month, equal-weighted	J = 6, 12 months; K = 5, 10	4	J = 12, K = 10

One month = 21 trading days. An EMA with span s uses smoothing weight $\alpha = 2/(s + 1)$. Families 1–7 are applied to each industry separately and then equal-weighted across industries. Family 8 selects across industries and is always fully invested, so it serves as a relative-trend comparison, not a crash hedge. It stays in the SPA test as a candidate that is unlikely to help; the SPA test is designed so that poorly performing candidates have little effect on its p-value. It is labeled in every table as a relative-trend rule rather than a hedge. Because it needs many assets to rank, family 8 is excluded from the single-market PPUT comparison and the 9-ETF check; those use flagships 1–7. Family 6 differs from Moreira & Muir, who scale by variance and allow leverage: it scales by volatility and caps the stock weight at 1. RSI, MACD, and Bollinger-band rules are excluded because they are short-term mean-reversion signals, not trend or risk-management rules.

Benchmarks:

- Buy-and-hold: equal-weighted industries, always fully invested, rebalanced daily, and charged no trading costs. This is deliberate and conservative against the strategy; a robustness run charges buy-and-hold the same cost c on its daily rebalancing turnover.
- Constant-exposure mix, one per flagship: a share $\bar{w}$ in stocks and $(1 - \bar{w})$ in T-bills. This isolates whether timing adds value beyond simply holding less stock. The primary version is ex-ante: each month, $\bar{w}$ is the flagship's average stock exposure over all prior months, so it uses no future information. Each flagship's first 12 evaluation months set the starting $\bar{w}$ and are excluded from the timing-value comparison. The ex-post version ($\bar{w}$ = full-sample average exposure) is a robustness check.

Flagship tests (secondary):

1. Each of the 8 flagships vs. buy-and-hold on ΔCE (the H1 metric). p-values adjusted with the Holm (1979) step-down correction across the 8 tests. Sharpe differences are reported alongside. The 50/200 flagship's unadjusted result is H1; Holm applies only to this 8-flagship comparison.
2. Each flagship vs. its constant-exposure mix on ΔCE , the timing-value test, also Holm-adjusted across the 8 tests. For family 8, which is always fully invested, the constant-exposure mix is buy-and-hold itself, so this test repeats test 1.
3. Crash-cycle returns and ΔCE at $\gamma = 2, 5, 10$ for all 8 flagships.

Comparison with explicit insurance (PPUT)

This test asks whether trend-following buys crash protection more cheaply than buying put options. It is a secondary test over the dates where PPUT and its S&P 500 total return base both

have data (1988–2025, since the S&P 500 total return series on yfinance starts on 1988-01-04).

The insurance benchmark: the Cboe S&P 500 5% Put Protection Index (PPUT). It holds the S&P 500 and buys a one-month put 5% below the market, rolled each month.

Matched comparison: PPUT is a single-index strategy on the S&P 500, so for this test flagships 1–7 are applied to the S&P 500 total return index instead of the industries, and buy-and-hold is that same index. Every series in this test shares one base, so the costs of protection are directly comparable. Family 8 is left out because a single index gives it nothing to rank. Crash cycles for this test are identified on the S&P 500 total return index with the same 20% rule.

Cost of protection, computed for each flagship and for PPUT relative to buy-and-hold (the S&P 500 total return index):

$$\text{Cost per point of drawdown protection} = \frac{R_{BH} - R_S}{|MDD_{BH}| - |MDD_S|}$$

R is the annualized return and MDD the maximum drawdown, both in percent. The same ratio is also reported with monthly 5% CVaR in place of MDD. A lower number means cheaper protection. Each ratio is only defined when the strategy actually reduces that risk measure; otherwise the strategy is reported as giving no protection.

PPUT is also compared to each flagship on crash-cycle returns (decline and recovery phases) and CRRA ΔCE at $\gamma = 5$, using the verdict bins defined under Economic significance and power.

Caveats stated in the paper: index puts are expensive on average because of the volatility risk premium (Coval & Shumway 2001), which works against PPUT. The comparison covers one index only, and the S&P 500 differs from the industry portfolios used in the main tests. PPUT values before the index’s launch are back-tested by Cboe rather than live.

Tests and inference

The paper’s headline result is one test: H1 (CRRA ΔCE , $\gamma = 5$) on the equal-weighted industry portfolio, full sample, 50/200 rule, 10 bp cost. Everything else is labeled secondary or exploratory in the paper.

Primary test (H1): mean of the monthly utility differences d_m , using a studentized stationary block bootstrap (Politis & Romano 1994) with 10,000 draws. Block length is chosen by the Politis-White (2004) automatic rule, as corrected by Patton, Politis & White (2009). The same bootstrap gives the 95% CI for ΔCE . A Newey-West t-test on d_m is reported as a check.

Secondary tests:

1. **Sharpe difference (H2):** Ledoit-Wolf (2008) studentized block bootstrap on daily excess returns.
2. **Mean excess return:** t-test on daily strategy-minus-buy-and-hold returns with Newey-West (1987) standard errors. Lag = floor($4(T/100)^{(2/9)}$).
3. **Factor alpha (H3):** regress daily strategy excess returns on FF3 + UMD (1926–2025) and FF5 + UMD (1963–2025), where UMD is the Carhart (1997) momentum factor, Newey-West errors.
4. **Market timing:** Treynor-Mazuy (1966) quadratic and Henriksson-Merton (1981) option-style regressions.
5. **Tail protection (H4):** one-sided bootstrap test of monthly CVaR at 5% (the H4 test); daily CVaR reported alongside. Max drawdown, Sortino, Calmar, and crash-cycle returns are reported descriptively with bootstrap CIs.
6. **Other risk levels:** ΔCE at $\gamma = 2$ and 10, plus mean-variance CE as a comparison.
7. **Publication decay (H5):** the interaction regression above, the bootstrap of ΔCE post minus pre, and subperiod Sharpes.

8. **Data snooping (H6):** Hansen (2005) SPA test on d_m over all 90 rules in the catalog, 10,000 bootstrap draws. White's (2000) Reality Check reported alongside.
9. **Multiple-testing hurdles:** Deflated Sharpe Ratio (Bailey & López de Prado 2014). Because the 90 rules are highly correlated, the effective number of trials is estimated by clustering rules on return correlation; results are reported at both the cluster count and $N = 90$. The $t > 3.0$ hurdle of Harvey, Liu & Zhu (2016) is reported alongside.
10. **Simulated null models (Brock-style):** run the 50/200 rule on 1,000 simulated series each from a bootstrapped random walk, an AR(1), and a GARCH(1,1) (Bollerslev 1986) fit to each industry. Report where actual results fall in the simulated distribution.
11. **Crash-episode dependence:** because $\gamma = 5$ gives a few extreme months most of the weight, H1 is rerun (a) excluding 1929–1939 and (b) leaving out each crash cycle in turn. The range of ΔCE across the leave-one-out runs is reported, showing whether any single episode drives the result.

Economic significance and power

Every result for the primary rule falls into one of four verdicts, set now, based on the annualized CRRRA ΔCE at $\gamma = 5$.

Verdict	Condition
Worth it, meaningful	$\Delta CE \geq 0.5\%$ per year and its 95% bootstrap CI excludes 0
Worth it, trivial	$\Delta CE > 0$ with CI excluding 0, but below 0.5% per year
Not worth it	$\Delta CE < 0$ with CI excluding 0
Inconclusive	CI includes 0; interpreted using the power analysis below

For H2, a Sharpe difference of at least 0.10 in either direction counts as economically meaningful. A statistically significant difference smaller than 0.10 is reported as trivial.

Power analysis (run on simulated data before any real test):

1. Build the equal-weighted industry buy-and-hold index. Resample its daily returns, with the matching daily T-bill rates, in blocks using a stationary bootstrap. Each resample is a synthetic price path as long as the real sample.
2. Run the 50/200 rule on each synthetic path to get synthetic strategy returns. No real strategy return is computed at any point, so the power analysis cannot reveal the real result.
3. Calibrate a constant shift once, on a single very long simulated path (100 times the sample length), so the synthetic strategy's population ΔCE equals $\Delta = 0.25, 0.5, 1.0, 1.5, 2.0\%$ per year. Apply that same shift to every synthetic path, so each path's estimated ΔCE varies around Δ the way a real sample would.
4. Run the H1 test on 500 independent synthetic paths at each Δ , using the Newey-West version for speed. Spot-check 50 runs per Δ with the full bootstrap.
5. The minimum detectable effect (MDE) is the smallest Δ rejected in at least 80% of runs at $\alpha = 0.05$.
6. Report the MDE separately for the full sample, pre-1993, and post-1992.

The power analysis works on the portfolio index rather than industry by industry; Appendix A states this approximation. Block lengths chosen by the Politis-White rule are short relative to a 200-day moving average, so synthetic paths understate long trends and volatility clustering, which are what the rule reacts to. Appendix A states this limitation, and the MDE is also reported with fixed block lengths of 252 and 1,260 trading days.

An inconclusive result with $MDE \leq 0.5\%$ per year is read as evidence of no meaningful effect. With $MDE > 0.5\%$ per year, it is read as the test being too weak to decide.

Return-drag decomposition: The strategy differs from buy-and-hold only on the part of the portfolio held in T-bills. Summing daily arithmetic returns, the gap splits exactly as:

$$\sum_t (r_t^S - r_t^{BH}) = \sum_t (1 - w_t) (r_{f,t} - r_t) - c \sum_t |w_t - w_{t-1}|$$

where w_t is the stock weight held on day t (0 or 1 for trend rules, fractional for vol targeting) and r_t is the industry's return. It is computed per industry and then averaged. The first sum is split into buckets, reported for flagships 1-7 (family 8 is always fully invested, so the identity does not apply to it):

- **Crash declines:** days in the decline phase of a crash cycle (peak → trough).
- **Crash recoveries:** days in the recovery phase (trough → back to the prior peak), where trend-following typically misses gains.
- **Avoided losses outside crash cycles:** other days in cash when the industry underperformed T-bills.
- **Missed rallies outside crash cycles:** other days in cash when the industry beat T-bills.
- **Trading costs:** the second term.

Whipsaws are also counted for on/off rules (families 1-5 and family 7's trend signal): an exit followed by re-entry within 20 trading days at a higher price. The paper reports their number and total contribution to the drag.

Code validation

The backtest engine must pass every check below before any real-data test runs. The test files live in the GitHub repo and run automatically on each commit.

Unit tests on synthetic data (exact match required):

Test	Expected result
Signal always long, starting invested (no entry trade)	Strategy returns equal buy-and-hold exactly
Signal always cash	Strategy returns equal the T-bill series exactly
Flat price series (trend rules, families 1-5)	Zero trades, zero cost
10-day hand-computed example with 2 trades	Matches hand calculation to 1e-10
Cost accounting	Total cost = $c \times \sum \Delta w $ exactly
Look-ahead check	Shuffling returns from day t onward does not change the position held on day t

Replication checks (small tolerance allowed, since data sources differ):

1. Compounded daily industry returns match French's own annual industry return file to within 0.1% per year. The tolerance allows for French rounding daily returns to 0.01%, which compounds to a few hundredths of a percent per year.
2. The Faber (2007) 10-month SMA rule on U.S. market data reproduces his Table 5 results for 1972-2005. He reports S&P 500 buy-and-hold at 11.24% CAGR and 17.47% standard deviation, and timing at 11.18% CAGR and 14.00% standard deviation ([paper](#)). For this check only, his conventions are used: total returns, trades at the month-end close, no trading costs. Because the French market is not the S&P 500, the check compares the timing-minus-buy-and-hold differences (−0.06 points of CAGR, −3.47 points of standard deviation), which must match within 0.5 percentage points. His cash rate (90-day commercial paper) differs from the T-bill rate, and any gap is explained in the paper. These figures come from the July 2006 working-paper version; they are checked against the published 2007 version before the replication runs.
3. Data validation: the first evaluation date is before 1929-09-01, so the 1929 crash is in the sample.

Sample splits and out-of-sample design

The split date is 1993-01-01, the first full year after Brock et al. (1992) was published.

Period	Dates	Purpose
Full sample	1926-07 to 2025-12	H1-H4, H6
Pre-publication	1926-07 to 1992-12	Rule selection for true out-of-sample test; H5 baseline
Post-publication	1993-01 to 2025-12	H5; out-of-sample evaluation
Investable era	1999-01 to 2025-12	ETF robustness check (flagships 1-7)

True out-of-sample test: Choose the single rule of the 90 with the highest ΔCE on 1926–1992 only. Apply it unchanged to 1993–2025 and compare it to the 50/200 rule and to buy-and-hold.

Rolling selection: Each January, re-pick the rule with the highest ΔCE using an expanding window of all prior data. Trade it for that year. This mimics a real investor who re-tunes with only past information. Rolling selection starts once 10 years of evaluation data exist.

Crash cycles (defined objectively, not by hand): every episode in which the market total-return index falls 20% or more from its prior peak. Each crash cycle has two phases: the decline (peak $\rightarrow$ trough) and the recovery (trough $\rightarrow$ back to the prior peak). Results are reported for each phase and the whole cycle, so the rebound that trend-following tends to miss is counted. A cycle still unrecovered at the end of the sample is reported as open. The list is generated by code and reported in the paper, so no episode is picked after seeing results.

International replication (appendix)

The monthly strategies are re-run on developed-market country portfolios as an out-of-sample check that the U.S. result is not a feature of U.S. history alone. This is supporting evidence, reported in an appendix, not a new hypothesis.

Data: French Data Library international country portfolios, monthly, in U.S. dollars, from 1975 to 2025-12, built from MSCI data through 2006 and Bloomberg data from 2007; the start date varies by country. Only each country's Mkt column from the "Value-Weight Dollar Returns, All 4 Data Items Not Req'd" block is used: a value-weighted return on all firms, in U.S. dollars. The value/growth sorts and the block requiring all four ratios are not used. This was confirmed in the downloaded file (Japan.Dat) on 2026-09-28. Countries are included if they have at least 30 years of data. The cash return is the U.S. T-bill rate.

Strategies: only the monthly flagships, since the data is monthly:

- Family 2: price vs. 10-month SMA (Faber)
- Family 4: 12-month time-series momentum

Costs are 10 bp one-way, as in the main sample.

What is reported, per country and pooled equal-weighted across countries:

- Sharpe difference vs. buy-and-hold, with bootstrap CI
- Max drawdown and 5% CVaR, strategy vs. buy-and-hold
- ΔCE at $\gamma = 5$ with its verdict bin
- Crash-cycle returns (decline and recovery phases), with cycles defined per country by the same 20%-drawdown rule

Reading the result: protection that appears in most countries supports a general effect. Protection only in the U.S. suggests the main result is country-specific. Lower power from monthly data is stated alongside every result.

Planned tables and figures

Each exhibit maps to one hypothesis, so nothing is added just because it looks good.

Exhibit	Content	Answers
Table 1	Descriptive statistics of industry returns, by period	Setup
Table 2	Performance of the 8 flagships, buy-and-hold, and constant-exposure mixes: return, vol, Sharpe, CRRA CE, MDD, CVaR, Sortino, Calmar, turnover, average exposure	H1, H2, H4
Table 3	Primary test: ΔCE ($\gamma = 5$) with 95% CI and verdict; Sharpe-difference and mean-excess tests; flagship ΔCE tests and timing-value tests with Holm-adjusted p-values	H1, H2
Table 4	Factor alphas and market-timing coefficients	H3
Table 5	Crash-cycle returns, decline and recovery phases, strategy vs. buy-and-hold	H4
Table 6	ΔCE at $\gamma = 2, 5, 10$ (CRRA) and mean-variance CE, with verdict bins; H1 excluding 1929-1939 and the leave-one-crash-out range	H1
Table 7	Pre- vs. post-1992 results, out-of-sample and rolling selection	H5
Table 8	SPA and Reality Check p-values, deflated Sharpe	H6
Table 9	Return-drag decomposition for flagships 1-7: crash declines, crash recoveries, avoided losses, missed rallies, costs, whipsaws	Mechanism
Table 10	Flagships 1-7 vs. PPUT, 1988-2025: cost per point of drawdown and CVaR protection, crash-cycle returns, ΔCE	Insurance comparison
Table 11	Cost sensitivity, break-even cost, and buy-and-hold charged its rebalancing costs	Robustness
Table 12	ETF-era replication, 1999-2025, flagships 1-7	Robustness
Figure 1	Equity curves, log scale, 1926-2025, crash cycles shaded	H1, H4
Figure 2	Drawdown profiles, strategy vs. buy-and-hold	H4
Figure 3	Rolling 10-year ΔCE with 1993 marked	H5
Figure 4	ΔCE heatmap across the 48-rule SMA grid, plus ΔCE distribution of all 90 rules by family	H6
Figure 5	Actual result vs. simulated null distributions	Secondary test 10
Figure 6	Protection bought vs. return given up: flagships 1-7 and PPUT plotted as points	Insurance comparison
Appendix A	Power analysis: rejection rate by true Δ , MDE per period and block length, with the block-length limitation stated	Test design
Appendix B	Code validation results: unit tests and replication checks	Credibility
Appendix C	Per-industry results for all 49 industries	Robustness
Appendix D	International replication: monthly flagships by country and pooled	External validity

Limitations and threats to validity

These go into the paper's limitations section with the mitigation for each.

Threat	Why it matters	Mitigation
Few independent crash episodes	With $\gamma = 5$, about ten crash cycles (and 1929-1932 above all) carry most of the weight in ΔCE	H1 excluding 1929-1939; leave-one-crash-out range
Prior exposure to the 50/200 rule	The capstone result shaped the framing and overlaps the post-1992 period in time	Disclosed under Research question; primary rule and tests fixed here

Threat	Why it matters	Mitigation
Industry portfolios not investable before ~1999	Results before ETFs are not directly achievable	ETF-era replication (Table 12)
Historical trading costs	Pre-1975 costs were far higher than 10 bp	Time-varying cost run; break-even cost
Residual delisting bias (Shumway 1997)	Some CRSP delisting returns are missing	Note as limitation; equal-weighted robustness
Daily rebalancing assumption	Unrealistic at scale	Monthly rebalancing robustness
Choice of risk-free rate	Affects Sharpe and cash return	Uses French's daily T-bill series, not an assumed constant
Choice of utility function	ΔCE depends on the CRRA form and on γ	Report $\gamma = 2, 5, 10$ and mean-variance CE alongside the primary $\gamma = 5$
Power analysis block length	Short bootstrap blocks understate long trends, so the MDE may be off	MDE also reported with 252- and 1,260-day blocks
Joint-hypothesis problem (Fama 1970)	Alpha depends on the chosen factor model	Report several factor models
Taxes ignored	Trading triggers realized gains	Stated limitation
Primary rule chosen by convention	50/200 is popular because of past performance	SPA over the full 90-rule catalog; deflated Sharpe

Locking rules

Once locked, the primary test and its metric (CRRA ΔCE , $\gamma = 5$), the rule catalog and flagships, sample splits, cost grid, crash-cycle definition, verdict thresholds, and α level do not change. Any change after locking goes in the deviation log with a date and reason, and the paper reports it.

- ☐ Review and edit this plan
- ☐ Download the French international country file and confirm it contains a whole-country market return in U.S. dollars (done 2026-09-28)
- ☐ Lock it: export a dated PDF of this plan
- ☐ Pre-register the PDF on OSF (osf.io) and save the registration link
- ☐ Commit the same PDF to the GitHub repo
- ☐ Only then run any real-data test (code validation on synthetic data may run anytime)

The paper's methods section will state: "The analysis plan was pre-registered on OSF on [date], before any results were computed," with the registration link.

Deviation log

Date	Change	Reason

Key references

All references below were checked online against publisher or index records (authors, year, title, journal, volume, issue, pages) on 2026-09-28.

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